Design of Imbalanced Mach–Zehnder Interferometers for Group-Index Extraction

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***Abstract*—**This proposal presents six imbalanced Mach–Zehnder interferometers (MZIs) designed to extract the group index of a 500×220 nm silicon-on-insulator strip waveguide for the quasi-TE mode, together with a grating-coupler loopback for insertion-loss calibration. The path-length difference ΔL is varied from 50.9 to 786.8 μm, producing free spectral ranges (FSRs) from approximately 11.7 to 0.75 nm near 1550 nm. Lumerical MODE simulations give an effective index $n_{\mathrm{eff}} = 2.446$ and a group index $n_g = 4.054$ at 1550 nm. A quadratic compact model is fitted to the simulated effective index and used to calculate the MZI transmission spectra, and the submitted layout is also simulated with the grating couplers included. After fabrication, the group index will be extracted from the measured fringe spacing and compared with the simulated value.

I. Introduction

THE group index n_g of a waveguide determines the free spectral range of resonators and interferometers and the delay of optical signals. An imbalanced MZI provides a direct way to measure it from the spacing between adjacent transmission fringes.

The objective of this design is to extract the group index of the standard strip waveguide from measured MZI spectra and compare it with the value obtained from waveguide simulation. Six MZIs with different path-length differences ΔL are included. For an ideal imbalanced MZI, increasing ΔL decreases the FSR, providing several devices with different fringe spacings for the same waveguide geometry. A loopback structure, consisting of two grating couplers connected by a waveguide, is also included to measure the grating-coupler response.

II. Theory

A. Waveguide compact model

The wavelength dependence of the effective index is represented by a second-order expansion around $\lambda_0 = 1.55$ μm [1]:

$$n_{\mathrm{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2. \tag{1}$$

The group index is defined as

$$n_g(\lambda) = n_{\mathrm{eff}}(\lambda) - \lambda \frac{dn_{\mathrm{eff}}}{d\lambda}. \tag{2}$$

For the polynomial in (1),

$$n_g(\lambda) = n_{\mathrm{eff}}(\lambda) - \lambda \left[n_2 + 2n_3(\lambda - \lambda_0) \right]. \tag{3}$$

B. MZI transfer function

In each MZI, a Y-branch divides the input field between two waveguide arms [2]. The arm lengths are L_1 and

$$L_2 = L_1 + \Delta L, \tag{4}$$

where ΔL is the path-length difference.

The propagation constant is

$$\beta(\lambda) = \frac{2\pi n_{\mathrm{eff}}(\lambda)}{\lambda}. \tag{5}$$

The phase difference accumulated between the two arms is therefore

$$\phi(\lambda) = \beta(\lambda)\Delta L = \frac{2\pi n_{\mathrm{eff}}(\lambda)\Delta L}{\lambda}. \tag{6}$$

Neglecting propagation loss, the normalized transmission for an MZI recombined by a second Y-branch, as used for MZI3–MZI6, is

$$T_Y = \frac{1}{2}\left[1 + \cos(\beta \Delta L)\right]. \tag{7}$$

MZI1 and MZI2 use a 2×2 3-dB coupler at the output. The two output ports have complementary transmission:

$$T_{1,2} = \frac{1}{2}\left[1 \pm \sin(\beta \Delta L)\right]. \tag{8}$$

C. Free spectral range and group-index extraction

Adjacent transmission minima or maxima occur when the interferometric phase changes by 2π. For a small wavelength interval corresponding to one FSR,

$$\left| \frac{d\phi}{d\lambda} \right| \mathrm{FSR} \approx 2\pi. \tag{9}$$

Differentiating (6) gives

$$\frac{d\phi}{d\lambda} = 2\pi \Delta L \frac{d}{d\lambda}\left(\frac{n_{\mathrm{eff}}}{\lambda}\right) \tag{10}$$

$$= \frac{2\pi \Delta L}{\lambda^2}\left(\lambda \frac{dn_{\mathrm{eff}}}{d\lambda} - n_{\mathrm{eff}}\right). \tag{11}$$

Using the definition of group index in (2),

$$\frac{d\phi}{d\lambda} = -\frac{2\pi n_g \Delta L}{\lambda^2}. \tag{12}$$

Substitution into (9) gives

$$\mathrm{FSR} \approx \frac{\lambda^2}{n_g \Delta L}. \tag{13}$$

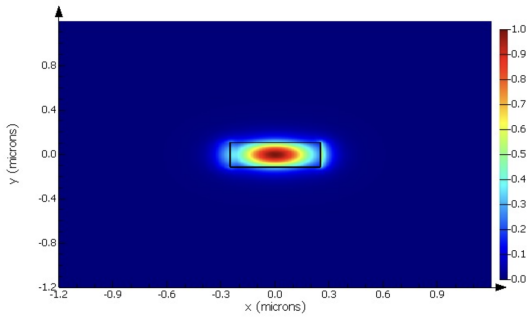


Fig. 1. Fundamental quasi-TE mode of the 500 × 220 nm strip waveguide at 1550 nm. The MODE simulation gives $n_{\mathrm{eff}} = 2.4463$, an effective area of 0.190 μm^2, and a TE polarization fraction of 98%.

The group index can therefore be obtained from the measured fringe spacing using

$$n_g \approx \frac{\lambda^2}{\Delta L \cdot \mathrm{FSR}}, \tag{14}$$

where λ is taken near the midpoint of the two adjacent fringes used to determine the FSR. Equation (14) assumes that the group index changes only slightly over one FSR.

III. Modelling and Simulation

A. Waveguide

The waveguide is a silicon strip with width $W = 500$ nm and height $H = 220$ nm on buried oxide with oxide cladding. The waveguide height is fixed by the fabrication process. Only the fundamental quasi-TE mode is considered, consistent with the TE grating couplers used in the layout.

The waveguide modes were calculated using the finite-difference eigenmode (FDE) solver in Lumerical MODE. The simulations used the Palik material models for Si and SiO_2 and a 2.5-nm mesh in both transverse directions. Figure 1 shows the fundamental quasi-TE mode at 1550 nm.

A wavelength sweep was performed at 10 points from 1550 to 1600 nm. Equation (1) was fitted to the simulated effective-index values. Figure 2 shows the MODE results and the fitted compact model.

With λ expressed in μm, the fitted model is

$$\begin{aligned} n_{\mathrm{eff}}(\lambda) = 2.4463 - 1.0374(\lambda - 1.55) \\ - 0.1329(\lambda - 1.55)^2. \end{aligned} \tag{15}$$

The maximum residual of the fit over the simulated wavelength range is 9×10^{-6}. Using (3), the compact model gives $n_g = 4.054$ at 1550 nm and $n_g = 4.075$ at 1600 nm. These values agree with the group-index values obtained directly from the MODE sweep. Values below 1550 nm in the circuit calculation use an extrapolation of the fitted polynomial.

B. Design variations

The arm lengths of the six MZIs were measured from the layout in KLayout using SiEPIC-Tools [3]. The resulting path-length differences are listed in Table I, together with the FSR near 1550 nm calculated from the compact model.

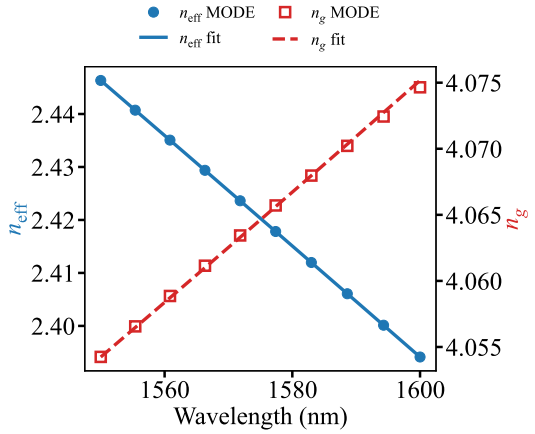


Fig. 2. Effective index and group index obtained from Lumerical MODE. The lines show the fitted effective-index model and the corresponding group index calculated from (3).

TABLE I
MZI Design Variations and Calculated FSR

Device	Combiner	ΔL (μm)	FSR (nm)
MZI5	Y-branch	50.877	11.70
MZI3	Y-branch	86.417	6.88
MZI1	2 × 2	109.237	5.43
MZI4	Y-branch	224.786	2.63
MZI6	Y-branch	418.786	1.42
MZI2	2 × 2	786.767	0.75

The six values of ΔL produce a range of measurable fringe spacings, from approximately 11.7 nm for MZI5 to 0.75 nm for MZI2.

The loopback consists of two grating couplers connected by a single 173-μm waveguide. Its transmission is set almost entirely by the two grating couplers, so it can be used to remove the grating-coupler response from the measured MZI spectra.

C. Circuit simulation

The FSR values in Table I were calculated using the wavelength-dependent effective index from (15) and the MZI transfer functions in (7) and (8), from the spacing between adjacent minima near 1550 nm.

The submitted layout was also simulated using the automated circuit simulation run on layout submissions (SiEPIC-Tools with the SiEPIC EBeam PDK compact models). These spectra include the grating couplers, the Y-branches, the directional couplers, and the waveguides as drawn. Figure 3 shows the simulated spectra of the six MZIs. MZI1 and MZI2 have two complementary output ports, while MZI3–MZI6 have a single monitored output. As expected, the fringe spacing decreases as ΔL increases.

Figure 4 shows the simulated loopback spectrum. The peak transmission is approximately -4.8 dB near 1554 nm, corresponding to about 2.4 dB per grating coupler, and the 3-dB bandwidth is approximately 48 nm.

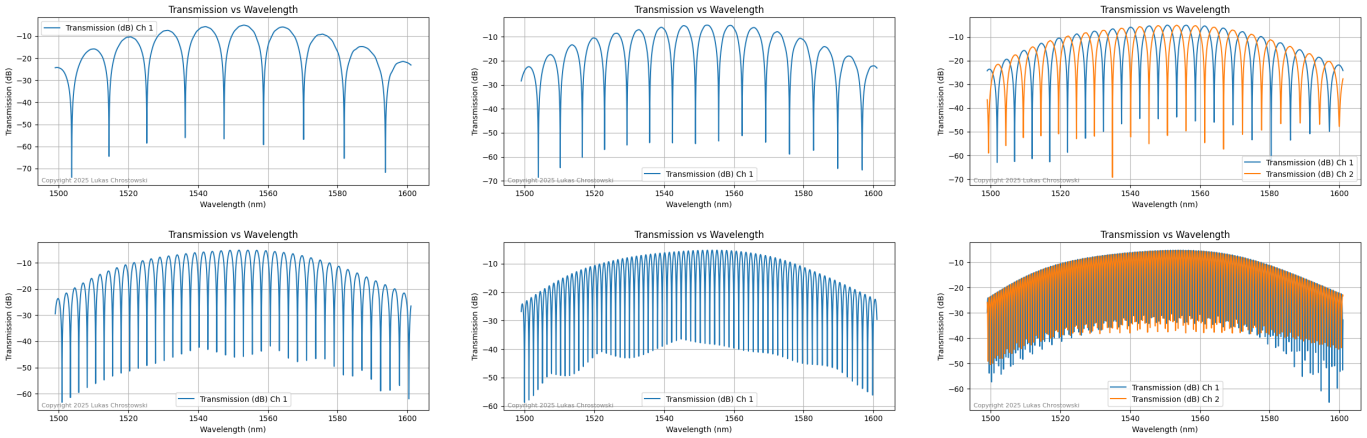


Fig. 3. Simulated transmission spectra of the submitted MZIs, including the grating couplers, in order of increasing ΔL. Top row, left to right: MZI5 ($\Delta L = 50.9$ μm), MZI3 (86.4 μm), and MZI1 (109.2 μm). Bottom row: MZI4 (224.8 μm), MZI6 (418.8 μm), and MZI2 (786.8 μm). MZI1 and MZI2 show both outputs of the 2×2 coupler.

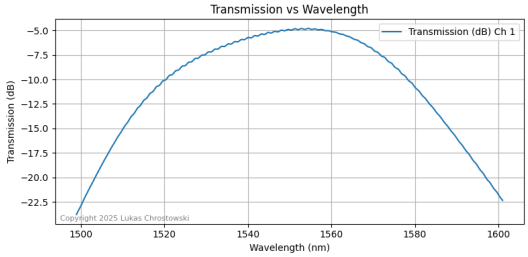


Fig. 4. Simulated transmission of the grating-coupler loopback.

D. Group-index extraction from simulated spectra

Equation (14) was first checked using the compact-model spectra. For MZI1, $\Delta L = 109.237$ μm and the calculated FSR is 5.43 nm near 1550 nm. Substituting these values into (14) gives $n_g \approx 4.05$, close to the MODE value of 4.054.

The same procedure was applied to the layout simulation. For MZI5, the FSR read from Fig. 3 is approximately 11.4 nm near 1553 nm, which gives $n_g \approx 4.17$; MZI3 gives a similar value. Because these values were read from the plotted spectra, they are accurate to about 2%. They are approximately 3% higher than the MODE value. The difference is attributed mainly to the silicon material model used in the MODE simulation, which shows little dispersion over this band.

E. Extraction from experimental data

After fabrication, the measured spectra will first be normalized using the loopback spectrum to remove the grating-coupler response. The FSR will then be measured from the spacing between adjacent minima in each MZI transmission spectrum. The group index will be calculated using (14) and the corresponding path-length difference ΔL, and compared with the simulations and across the six MZI designs.

IV. Fabrication

To be completed later.

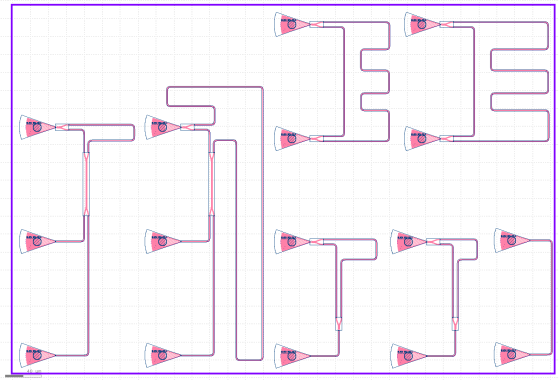


Fig. 5. Layout of EBeam_vaddiyaswant.gds. Left to right: MZI1 and MZI2; top row: MZI4 and MZI6; bottom row: MZI3, MZI5, and the loopback.

The submitted layout is shown in Fig. 5. Each device uses TE grating couplers placed on the standard 127-μm pitch for automated optical testing. MZI1 and MZI2 use one input and two output grating couplers, while MZI3–MZI6 and the loopback use one input and one output grating coupler.

V. Experimental Data

To be completed later.

VI. Analysis

To be completed later.

VII. Conclusion

To be completed later.

References

[1] L. Chrostowski and M. Hochberg, *Silicon photonics design: from devices to systems*. Cambridge University Press, 2015.

[2] Y. Zhang, S. Yang, A. E.-J. Lim, G.-Q. Lo, C. Galland, T. Baehr-Jones, and M. Hochberg, "A compact and low loss y-junction for submicron silicon waveguide," *Optics express*, vol. 21, no. 1, pp. 1310–1316, 2013.

[3] "SiEPIC-Tools: KLayout package for silicon photonics layout and verification," https://github.com/SiEPIC/SiEPIC-Tools.